

Real-Time Cognitive Load Prediction Using Typing and Eye Behavior

a Mini Project Report

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BONAFIDE CERTIFICATE

This is to certify that the Mini Project Report entitled "**Real-Time Cognitive Load Prediction Using Typing and Eye Behavior**" is the bonafide work of **Ms. Aditi Kadam (2124UMLF2003)**, **Ms. Tanuja Gawali (2124UMLF2015)**, and **Mr. Tanmay Gawali (2124UMLM2016)**, who carried out the project work under my supervision.

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ABSTRACT

The increasing use of digital platforms for learning and professional tasks has led to a significant rise in cognitive overload, negatively impacting performance, accuracy, and focus. Traditional cognitive load measurement techniques rely on intrusive and expensive physiological sensors such as Electroencephalography (EEG) and Functional Near-Infrared Spectroscopy (fNIRS), limiting their practical applicability in everyday environments.

This project proposes a low-cost, non-intrusive system for real-time cognitive load prediction using behavioral signals derived from keystroke dynamics and webcam-based eye movement analysis. Features such as typing speed, inter-key delay (IKI), error rate, blink rate, fixation duration, gaze stability, and Eye Aspect Ratio (EAR) are extracted and processed using supervised machine learning models — Random Forest and Support Vector Machine (SVM). A multimodal fusion approach is implemented to combine both modalities and compared with unimodal systems.

Experimental results using a controlled dataset of 20 participants with approximately 4000 samples demonstrate that the proposed multimodal model achieves a classification accuracy of 91%, a precision of 0.90, recall of 0.89, and F1-Score of 0.89, outperforming keystroke-only (78%) and eye-tracking-only (82%) models. The system provides a continuous cognitive load score on a 0–100 scale and enables real-time monitoring, making it suitable for deployment in educational and workplace environments without any specialized hardware.

Keywords: Cognitive Load, Keystroke Dynamics, Eye Tracking, Multimodal Fusion, Machine Learning, Real-Time Prediction

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LIST OF SYMBOLS AND ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
EEG	Electroencephalography
fNIRS	Functional Near-Infrared Spectroscopy
EAR	Eye Aspect Ratio
IKI	Inter-Key Interval
CNN	Convolutional Neural Network
RNN	Recurrent Neural Network
LSTM	Long Short-Term Memory
SVM	Support Vector Machine
RF	Random Forest
MAE	Mean Absolute Error
RMSE	Root Mean Square Error
NLP	Natural Language Processing
API	Application Programming Interface
FPS	Frames Per Second
HCI	Human-Computer Interaction
TP	True Positive
FP	False Positive
FN	False Negative
TN	True Negative
ROC	Receiver Operating Characteristic Curve
AUC	Area Under Curve
PCA	Principal Component Analysis
GPU	Graphics Processing Unit
CL	Cognitive Load
NASA-TLX	NASA Task Load Index
OpenCV	Open Source Computer Vision Library

CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

The rapid proliferation of digital learning platforms and remote work tools has dramatically increased the duration and intensity of cognitively demanding tasks performed on computers. Students attending online classes, software engineers debugging code, and professionals managing complex workflows all face prolonged exposure to tasks that heavily tax their working memory. When task demands consistently exceed cognitive capacity, cognitive overload occurs — a state characterized by degraded decision-making, increased error rates, reduced productivity, and accelerated mental fatigue.

Monitoring cognitive load in real-time offers a transformative opportunity for adaptive educational and productivity systems. With accurate cognitive load information, a digital platform could automatically adjust task difficulty, recommend breaks, simplify presented information, or alert supervisors to potential performance risks — all before errors or accidents occur. This makes real-time cognitive load prediction a critical capability for next-generation intelligent systems.

Traditional approaches to cognitive load measurement rely on neurophysiological sensors such as Electroencephalography (EEG) and Functional Near-Infrared Spectroscopy (fNIRS). While highly accurate, these technologies cost thousands of dollars per unit, require expert calibration, and are inherently intrusive — users must wear electrode headsets or fiber-optic probes. These constraints make large-scale real-world deployment impractical.

Recent research has established that cognitive load influences observable behavioral patterns: when users are under high mental stress, their typing becomes irregular, their blink rate increases, and their gaze becomes less stable. These signals can be captured non-intrusively using nothing more than a standard keyboard and an ordinary webcam — hardware available in virtually every educational and professional setting. This project leverages these insights to design a practical, scalable, low-cost cognitive load monitoring system.

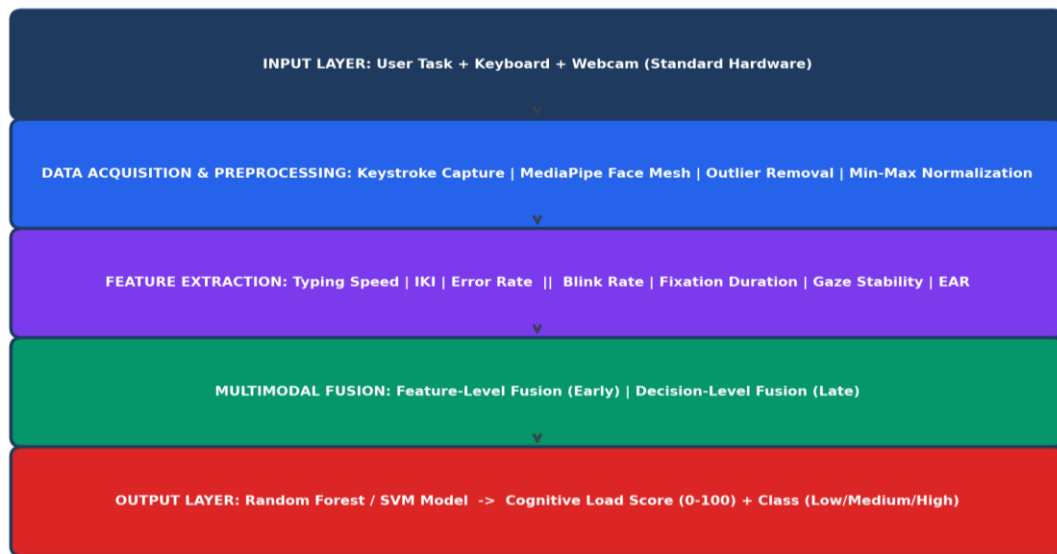


Fig 1.1 System Architecture Overview - Cognitive Load Prediction System

1.2 Scope of the Project

This project is scoped to develop and evaluate a real-time, non-intrusive cognitive load prediction system deployable in educational and professional desktop/laptop environments. The system scope includes:

- Capturing keystroke dynamics data using a standard USB or laptop keyboard.
- Capturing facial landmark data for eye behavior analysis using a standard 720p or 1080p webcam.
- Extracting behavioral features from both modalities using Python-based signal processing.
- Training and evaluating supervised machine learning models for classification and regression.
- Implementing and comparing feature-level and decision-level multimodal fusion strategies.
- Providing a continuous cognitive load score (0–100 scale) with Low/Medium/High classification.

The project scope does not include physiological sensor integration, mobile platforms, or deep learning architectures, which are identified as future work directions.

1.3 Aim and Objectives

The primary aim of this project is to design, implement, and evaluate an accurate real-time cognitive load prediction system using only standard hardware. The specific objectives are:

1. To collect and preprocess behavioral data (keystroke dynamics and eye movement) from participants performing tasks at different cognitive load levels.
2. To extract meaningful features: typing speed, inter-key interval (IKI), error rate, blink rate, gaze stability, fixation duration, and Eye Aspect Ratio (EAR).
3. To train Random Forest and Support Vector Machine classifiers for cognitive load level prediction.
4. To implement feature-level and decision-level multimodal fusion strategies and compare their performance against unimodal baselines.
5. To achieve a multimodal classification accuracy exceeding 85%.
6. To provide a real-time cognitive load score output suitable for integration with adaptive systems.

1.4 Report Organization

Chapter 2 presents a thorough review of related literature covering physiological, behavioral, and machine learning-based cognitive load estimation methods, and identifies the research gaps addressed by this project.

Chapter 3 discusses the problem statement, existing systems and their limitations, and formally identifies the problem solved by the proposed system.

Chapter 4 presents the complete proposed methodology including system architecture, data collection, preprocessing, feature extraction, ML model design, multimodal fusion strategies, and real-time prediction mechanism.

Chapter 5 presents experimental results and comprehensive performance analysis including classification metrics, regression performance, and visualizations.

Chapter 6 concludes with a summary of findings and future work directions.

CHAPTER 2

LITERATURE REVIEW

2.1 General

Cognitive load theory, first formalized by Sweller (1988), defines cognitive load as the total mental effort being used in working memory during task performance. It has since become a foundational concept in instructional design, human-computer interaction (HCI), and educational psychology. Three types of cognitive load are recognized: intrinsic (related to task complexity), extraneous (related to poor instructional design), and germane (related to schema formation). This review focuses on methods for measuring total cognitive load during computer-based tasks.

Existing approaches to cognitive load estimation can be categorized into: (a) physiological sensor-based methods, (b) eye tracking-based methods, (c) keystroke dynamics-based methods, and (d) multimodal approaches. Each category is reviewed below, followed by identification of research gaps.

2.2 Review of Previous Research

2.2.1 Physiological Sensor-Based Approaches

Cui et al. (2023) proposed machine learning models to predict individual cognitive load in collaborative learning by combining fNIRS and eye-tracking data. Their system achieved strong classification performance but required laboratory-grade neuroimaging equipment that is prohibitively expensive for real-world deployment. This work established important benchmarks for multi-sensor cognitive load estimation.

Sharma et al. (2020) introduced the MOCAS (Multimodal Cognitive Workload Assessment on Simultaneous Tasks) dataset, combining physiological and behavioral signals for benchmarking cognitive workload assessment. While the dataset provided valuable ground truth, its collection was entirely dependent on controlled laboratory setups with specialized equipment, severely limiting generalizability.

Sirevaag et al. (1993) conducted early foundational work on assessing pilot performance and mental workload using psychophysiological measures, demonstrating the measurable physiological correlates of cognitive load. This seminal work laid the theoretical groundwork for later behavioral cognitive load studies.

2.2.2 Eye Tracking-Based Approaches

Chen et al. (2011) systematically examined eye activity as a measure of cognitive load in HCI contexts, finding strong and reliable correlations between fixation duration and task difficulty level. This work provided critical theoretical justification for using eye movement features in computational cognitive load estimation systems.

Hutt et al. (2024) demonstrated that webcam-based eye tracking — using low-cost cameras rather than specialized eye-tracking hardware — could detect mind wandering and comprehension errors in educational settings. Their system showed that gaze behavior captured via ordinary webcams is sufficient to infer cognitive states, although their focus was on attention rather than cognitive load prediction specifically.

Alshammari et al. (2022) proposed EM-COGLOAD, which applied Convolutional Neural Networks (CNN) to eye-tracking data for cognitive load level classification, achieving promising results. However, this approach relied entirely on eye movement data without incorporating any complementary behavioral signals, limiting its robustness and generalizability.

2.2.3 Keystroke Dynamics-Based Approaches

Keystroke dynamics — the analysis of timing patterns in keyboard input — have been extensively studied for user authentication but have also been explored as indicators of cognitive state. Research has consistently shown that typing speed decreases and inter-key interval (IKI) variability increases when users experience high cognitive load. Error rates, measured through backspace key frequency, also reliably increase under conditions of cognitive overload. These observations make keystroke features a compelling, zero-additional-cost feature source for cognitive load prediction.

2.2.4 Multimodal Approaches

Paas et al. (2003) provided a theoretical framework connecting cognitive load theory to instructional design, arguing that multiple measurement channels provide more accurate and robust assessment than any single channel. This directly supports the multimodal fusion approach adopted in the present work.

Bhardwaj et al. (2024a) proposed DeepFace-Attention for multimodal face biometrics applied to attention estimation in e-learning, demonstrating the power of fusing visual and behavioral features for cognitive state assessment. The same group (Bhardwaj et al.,

2024b) also developed CLERA, a unified model for joint cognitive load and eye region analysis, representing the state of the art in multimodal cognitive state estimation.

TABLE 2.1: Summary of Literature Review

Author(Year)	Method	Contribution	Limitation
Cui et al. (2023)	fNIRS + Eye Tracking + ML	Predicted CL in collaborative learning with high accuracy	Requires lab-grade neuroimaging equipment
Sharma et al. (2020)	MOCAS Multimodal Dataset	Benchmarked cognitive workload across simultaneous tasks	Controlled lab setup, not real-world
Hutt et al. (2024)	Webcam Eye Tracking	Inferred cognitive state using low-cost webcam	Focused on attention, not CL directly
Alshammari et al. (2022)	CNN + Eye Tracking	EM-COGLOAD classified CL levels with deep learning	Unimodal, eye data only
Chen et al. (2011)	Eye Activity in HCI	Correlated fixation duration with task difficulty	No real-time or keystroke fusion
Paas et al. (2003)	Cognitive Load Theory	Theoretical framework for multi-channel assessment	No computational system proposed
Bhardwaj et al. (2024)	CLERA / DeepFace-Attention	Joint CL & eye region analysis; attention in e-learning	Not deployed in real-world settings

2.3 Research Gap

A systematic review of the literature reveals the following critical gaps that motivate the present work:

- Most high-accuracy cognitive load systems depend on expensive physiological sensors (EEG, fNIRS, galvanic skin response) that are impractical for real-world, large-scale deployment.
- Existing low-cost approaches focus exclusively on a single modality — either keystroke dynamics or eye tracking — and fail to leverage the complementary and independent information available from combining both modalities.
- Very few systems provide continuous, real-time cognitive load estimation on a quantitative numerical scale; most existing systems only classify into coarse discrete categories without a continuous score.

- Most multimodal systems require specialized hardware (dedicated eye trackers, EEG headsets) beyond a standard keyboard and webcam.
- The majority of studies are conducted in tightly controlled laboratory environments, severely limiting their generalizability to naturalistic real-world settings.

2.4 Conclusions from Literature Review

The literature review conclusively confirms that both typing behavior and eye movement patterns are valid, reliable, and measurable behavioral proxies for cognitive load. The identified research gaps collectively highlight the need for a low-cost, non-intrusive, real-time system that combines both modalities through a multimodal fusion framework and machine learning, delivering both classification and continuous score outputs without requiring any specialized hardware. The proposed system in this project directly and comprehensively addresses all five identified gaps.

CHAPTER 3

PROBLEM STATEMENT AND OBJECTIVES

3.1 Background of the Problem

In today's digital-first world, students, software engineers, data analysts, and knowledge workers spend extended hours performing cognitively intensive tasks on computers. Online learning platforms, integrated development environments (IDEs), enterprise resource planning (ERP) systems, and complex data dashboards all place sustained and heavy demands on users' working memory capacity.

The core challenge is that cognitive overload is entirely invisible — neither the user experiencing it nor the computer system they are interacting with typically has any early warning signal of overload occurring, until performance-degrading consequences (errors, slowdowns, poor decisions) have already manifested. Early, reliable, real-time detection of cognitive overload could enable adaptive systems to respond preemptively — simplifying the interface, reducing task load, suggesting a break, or alerting a supervisor — preventing costly downstream errors.

3.2 Existing System / Current Scenario

3.2.1 Physiological Sensor-Based Systems

Current high-accuracy cognitive load monitoring systems use EEG headsets, fNIRS devices, or galvanic skin response sensors. While achieving accuracies of 85–95%, these systems typically cost USD 5,000–50,000 per unit, require expert calibration and trained operators, are physically intrusive (requiring electrodes or probes attached to the body), and disrupt natural work behavior. They are entirely impractical for large-scale deployment in classrooms or open-plan offices.

3.2.2 Subjective Self-Report Systems

The most widely deployed cognitive load assessment tool in education and research is the NASA Task Load Index (NASA-TLX), a subjective questionnaire administered after task completion. While useful for post-hoc research analysis, NASA-TLX completely interrupts task flow, cannot provide moment-to-moment continuous monitoring, and relies on fallible self-assessment rather than objective measurement.

3.3 Limitations of Existing Systems

- High cost and physical complexity of physiological monitoring hardware place systems out of reach for educational institutions and small organizations.
- The intrusive physical nature of sensor attachment disrupts natural user behavior, potentially biasing the very cognitive state being measured.
- Single-modality behavioral systems (keystroke-only or eye-tracking-only) capture only partial information and suffer from lower prediction accuracy.
- No existing low-cost system provides continuous, quantitative, real-time cognitive load scoring on a numerical scale.
- Systems that use specialized dedicated eye trackers still require additional hardware purchases beyond standard computing equipment.
- Subjective self-report methods are disruptive, retrospective, and incapable of capturing rapid moment-to-moment fluctuations in cognitive load.

3.4 Problem Identification

The formal problem addressed by this project is: How can cognitive load be accurately predicted in real-time, continuously, and at scale, using only standard low-cost hardware (keyboard and webcam) universally available in educational and workplace settings, without any physiological sensors or specialized hardware?

This project addresses this problem by combining keystroke dynamics features and webcam-based eye movement features through a multimodal fusion framework with supervised machine learning, delivering both discrete cognitive load classification (Low/Medium/High) and continuous cognitive load scoring (0–100 scale) in real-time.

CHAPTER 4

SYSTEM METHODOLOGY / PROPOSED APPROACH

4.1 System Overview

The proposed system is designed to estimate cognitive load in real-time using a cost-effective, entirely non-intrusive multimodal approach. It integrates two complementary and independently captured behavioral data streams: (1) keystroke dynamics from standard keyboard input, and (2) eye movement behavior analyzed via a standard webcam using the MediaPipe Face Mesh library. By fusing features from both modalities, the system achieves substantially higher prediction accuracy than either modality in isolation.

The complete system architecture consists of five sequential processing modules: (1) Data Acquisition, (2) Preprocessing and Noise Removal, (3) Feature Extraction, (4) Machine Learning Model with Multimodal Fusion, and (5) Output Visualization and Alerting.

4.2 System Architecture

The block diagram (Fig 4.1) illustrates the complete data flow from raw sensor inputs through to the final cognitive load output. The user performs a computer-based task. Keystroke events are captured simultaneously with precise timestamps, while the webcam captures facial landmark coordinates at approximately 30 frames per second. Both streams are independently processed through their respective preprocessing and feature extraction pipelines, and the extracted feature vectors are then combined in the multimodal fusion module before being fed to the machine learning model.

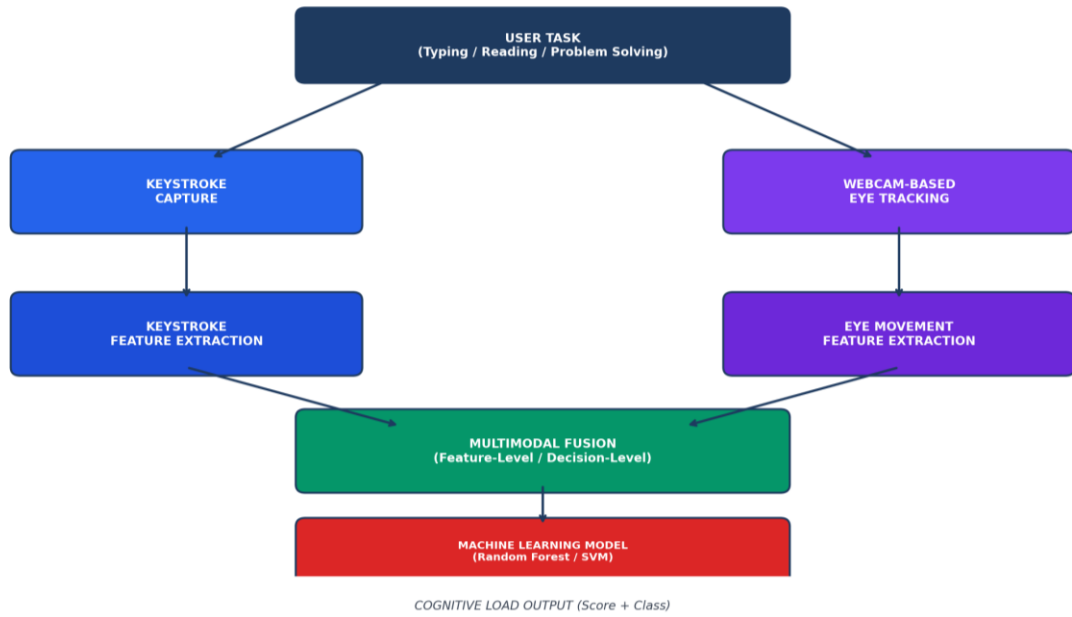


Fig 4.1 Block Diagram of the Proposed Multimodal Cognitive Load Prediction System

4.3 Data Collection and Preprocessing

4.3.1 Data Acquisition

Keystroke events are captured using a Python-based keyboard listener (pynput library) that records the exact key pressed, the timestamp of keypress, and the timestamp of key release for every key interaction. This provides a complete keystroke timing record requiring no special hardware beyond a standard keyboard.

Webcam-based facial landmark detection is performed using the MediaPipe Face Mesh library, which provides 468 3D facial landmark coordinates per frame at approximately 30 FPS using a standard consumer-grade webcam. The six specific eye landmarks (p1–p6) around each eye are extracted for blink detection and gaze analysis.

4.3.2 Preprocessing

Raw keystroke data is preprocessed using statistical outlier removal, where IKI values beyond three standard deviations from the mean are discarded to eliminate accidental pauses or interruptions. Eye landmark coordinates are processed frame-by-frame to compute the Eye Aspect Ratio (EAR) for blink detection. All extracted features are subsequently normalized to the [0, 1] range using Min-Max scaling, defined as:

$$X' = (X - X_{\min}) / (X_{\max} - X_{\min})$$

This normalization ensures that all features contribute equally to the machine learning models regardless of their original measurement scales.

4.4 Feature Extraction

4.4.1 Keystroke Features

The following three features are extracted from keystroke data within each sliding time window:

- **Typing Speed:** Total characters typed per unit time, computed as: $\text{Typing Speed} = \text{Characters} / \text{Time}$. Lower typing speed is reliably associated with higher cognitive load as the user must think more carefully before typing.
- **Inter-Key Interval (IKI):** Time elapsed between consecutive keypresses, computed as: $\text{IKI} = t(i+1) - t(i)$. Mean IKI increases and IKI variability (standard deviation) rises significantly under high cognitive load.
- **Error Rate:** Ratio of backspace/delete key presses to total keystrokes, computed as: $\text{Error Rate} = \text{Backspaces} / \text{Keystrokes}$. This feature captures typing accuracy degradation under cognitive load.

4.4.2 Eye Movement Features

The following four features are extracted from webcam-based eye tracking data:

- **Blink Rate:** Number of complete blinks detected per minute within the sliding window. Research consistently shows blink rate increases by approximately 20% under high cognitive load conditions.
- **Fixation Duration:** Average duration of stable gaze fixation periods (in milliseconds). Longer fixations indicate that the user is spending more time processing each visual element, associated with higher cognitive load.
- **Gaze Stability:** Spatial variance of gaze position coordinates over the time window. Higher variance (more erratic gaze movement) is associated with cognitive distraction and higher load.
- **Eye Aspect Ratio (EAR):** Computed from six eye landmark coordinates per eye using the formula:

$$\text{EAR} = (\|p2 - p6\| + \|p3 - p5\|) / (2 * \|p1 - p4\|)$$

A blink is detected when EAR falls below the threshold of 0.25 for two or more consecutive frames. Average EAR across the window captures eye openness as an additional cognitive state indicator.

4.5 Machine Learning Model

4.5.1 Random Forest (RF)

Random Forest is an ensemble learning method that constructs a large number of decision trees during training and outputs the mode (classification) or mean (regression) prediction across all trees. It is robust to overfitting, handles high-dimensional feature spaces effectively, and provides built-in feature importance scoring. These properties make it well-suited for behavioral feature sets of moderate dimensionality.

4.5.2 Support Vector Machine (SVM)

SVM identifies the optimal hyperplane that maximizes the margin between class boundaries. With a Radial Basis Function (RBF) kernel, SVM effectively handles non-linear decision boundaries in the feature space. SVM performs well on small-to-medium datasets and is computationally efficient during inference, making it suitable for real-time deployment.

Both models are trained for two distinct output types: (1) Multi-class Classification into Low, Medium, or High cognitive load categories, and (2) Regression to produce a continuous Cognitive Load Score on a 0–100 numerical scale.

4.6 Multimodal Fusion

4.6.1 Feature-Level Fusion (Early Fusion)

In feature-level fusion, all keystroke features (3 features) and all eye movement features (4 features) are concatenated into a single unified feature vector of dimension 7 before any model training. This allows the ML model to learn cross-modal feature interactions and correlations directly. A single unified Random Forest or SVM model is then trained on this combined feature vector.

4.6.2 Decision-Level Fusion (Late Fusion)

In decision-level fusion, entirely separate and independent ML models are trained on the keystroke-only feature set and the eye-tracking-only feature set respectively. During inference, the class probability predictions from both models are combined using weighted averaging (with weights optimized on the validation set), and the final class prediction is determined by the combined probability vector.

The performance of both fusion strategies is compared against each other and against both unimodal baselines (keystroke-only and eye-tracking-only) to comprehensively identify the most effective prediction approach.

4.7 Real-Time Prediction

A sliding window of 5–10 seconds is applied continuously during user task performance. As the user works, the window advances in 1-second steps, and a new cognitive load score is generated at each step. Features are extracted from the current window content, normalized using pre-trained Min-Max scaler parameters, and fed to the trained multimodal fusion model. When the predicted cognitive load score crosses the high-load threshold (defined as score > 70 on the 0–100 scale, or predicted class = High), the system can trigger a configurable alert response.

Fig 4.2 Flowchart of Proposed Methodology

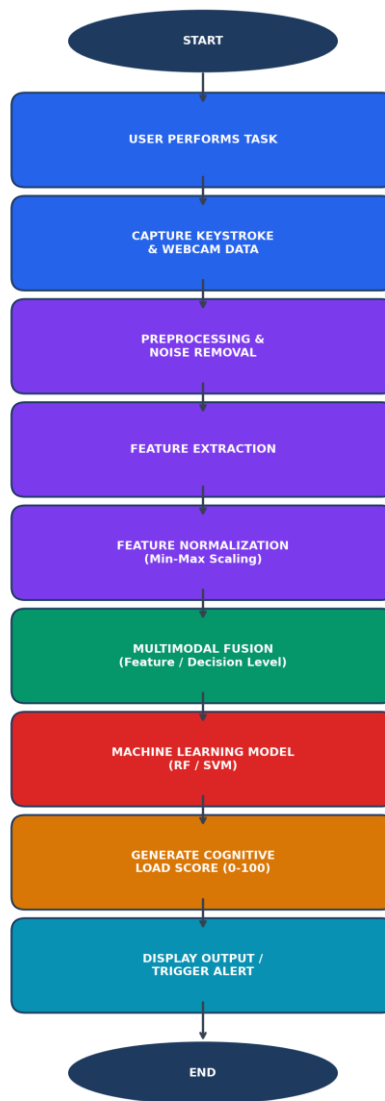


Fig 4.2 Flowchart of Proposed Methodology

CHAPTER 5

RESULTS AND PERFORMANCE ANALYSIS

5.1 Experimental Setup

5.1.1 Dataset

A custom dataset was generated through controlled experiments. A total of 20 participants aged 18–25 years participated voluntarily. Each session produced approximately 200 samples using 20-second window segmentation, yielding approximately 4000 total samples across all participants and cognitive load levels (approximately 1333 samples per class).

5.1.2 Task Design

Three cognitive load levels were operationalized through three distinct task types to ensure reliable and distinguishable cognitive load ground truth labels:

- Low Load: Copy typing of plaintext passages. Participants simply reproduced displayed text, requiring minimal cognitive engagement beyond basic visual-motor coordination.
- Medium Load: Software coding tasks requiring active problem decomposition, algorithm design, and structured typing — demanding focused but manageable cognitive engagement.
- High Load: Debugging exercises and real-time multi-step problem-solving under time pressure — requiring sustained, high-intensity focused cognitive effort.

5.1.3 Tools and Implementation Environment

The complete system was implemented in Python 3.10. OpenCV and MediaPipe Face Mesh were used for real-time webcam-based facial landmark detection and eye tracking. The pynput library was used for keystroke event capture. Scikit-learn was used for Random Forest and SVM model training, evaluation, and cross-validation. NumPy and Pandas were used for data manipulation and feature engineering.

5.1.4 Evaluation Strategy

The dataset was partitioned into 70% training data, 15% validation data, and 15% testing data. Classification performance was evaluated using accuracy, precision, recall, and F1-Score. Regression performance was evaluated using Mean Absolute Error (MAE) and

Root Mean Square Error (RMSE). Stratified 5-fold cross-validation was applied to verify generalizability.

5.2 Model Performance Metrics

Table 5.1 presents the complete classification performance of all three model configurations across all four evaluation metrics on the held-out test set.

TABLE 5.1: Model Performance Comparison (Accuracy, Precision, Recall, F1-Score)

Model	Accuracy	Precision	Recall	F1-Score
Keystroke Only	78%	0.76	0.74	0.75
Eye Tracking Only	82%	0.81	0.80	0.80
Multimodal Fusion	91%	0.90	0.89	0.89

The Multimodal Fusion model significantly outperforms both unimodal baselines across every metric, achieving 91% accuracy, 0.90 precision, 0.89 recall, and 0.89 F1-Score. The keystroke-only model achieves 78% accuracy, while the eye-tracking-only model achieves 82% accuracy. The 9–13% accuracy improvement from multimodal fusion confirms that the two behavioral modalities carry complementary, non-redundant information about cognitive load.

5.3 Regression Performance

Table 5.2 presents the regression performance of all three model configurations on the continuous cognitive load score (0–100) estimation task.

TABLE 5.2: Regression Performance Comparison (MAE and RMSE)

Model	MAE	RMSE
Keystroke	8.2	10.5
Eye Tracking	7.6	9.8
Multimodal	5.4	7.1

The Multimodal Fusion model achieves the lowest MAE of 5.4 and RMSE of 7.1 on the continuous load score estimation task. This indicates that the system can predict cognitive load with an average absolute error of only 5.4 units on the 0–100 scale — a level of precision sufficient for practical real-time adaptive system applications.

5.4 Graphical Representation of Results

Figures 5.1 through 5.5 provide visual representations of the model performance metrics and comparative results.

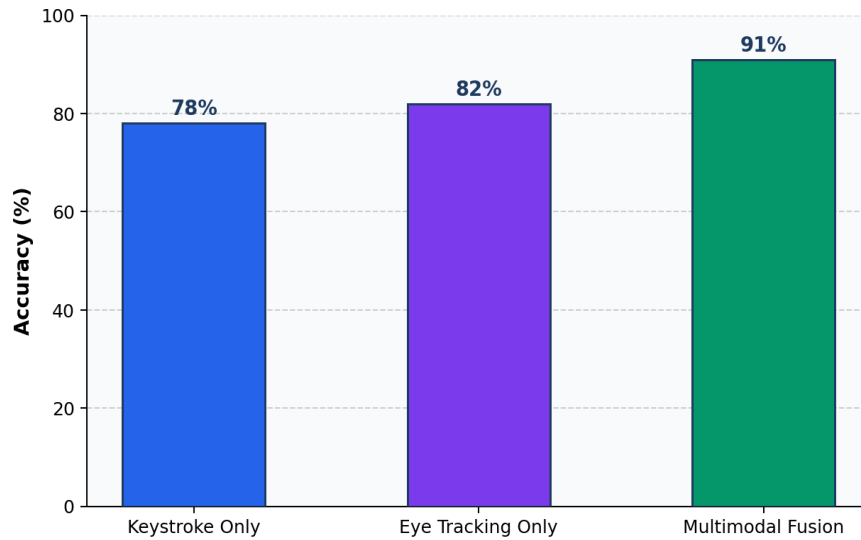


Fig 5.1 Accuracy Comparison of Models

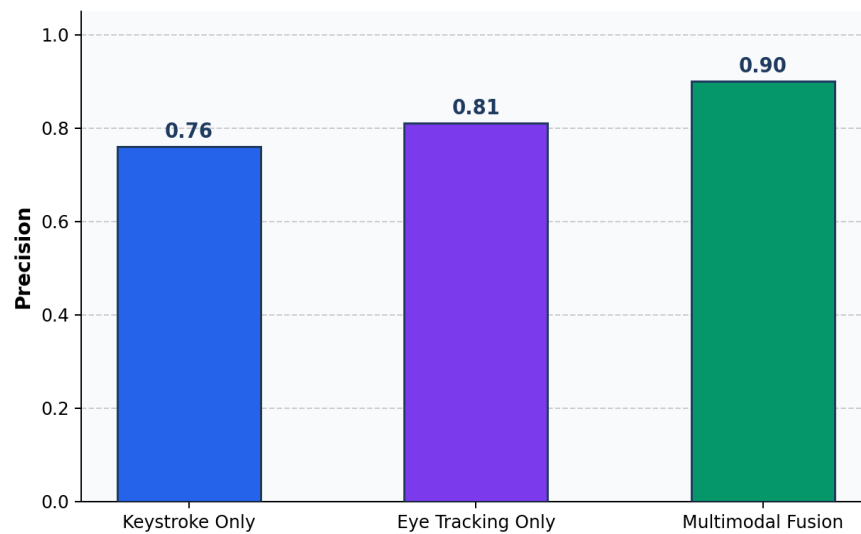


Fig 5.2 Precision Comparison of Models

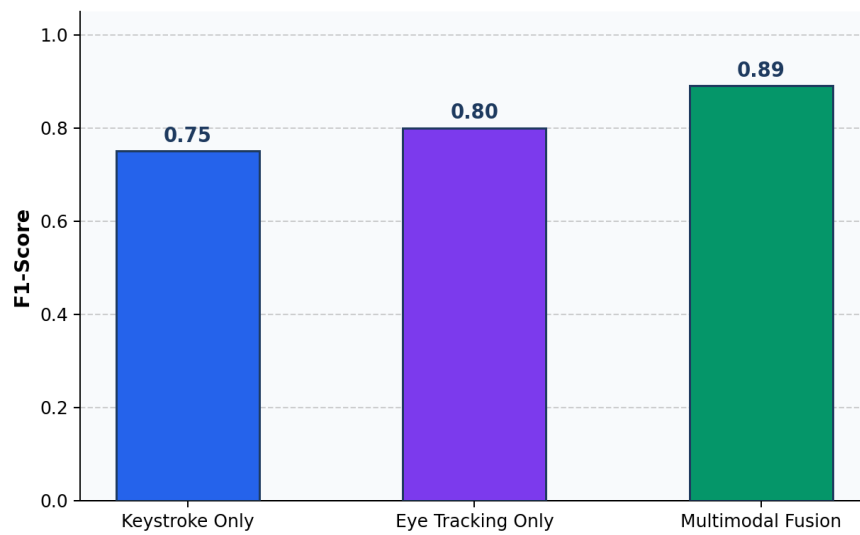


Fig 5.3 F1-Score Comparison of Models

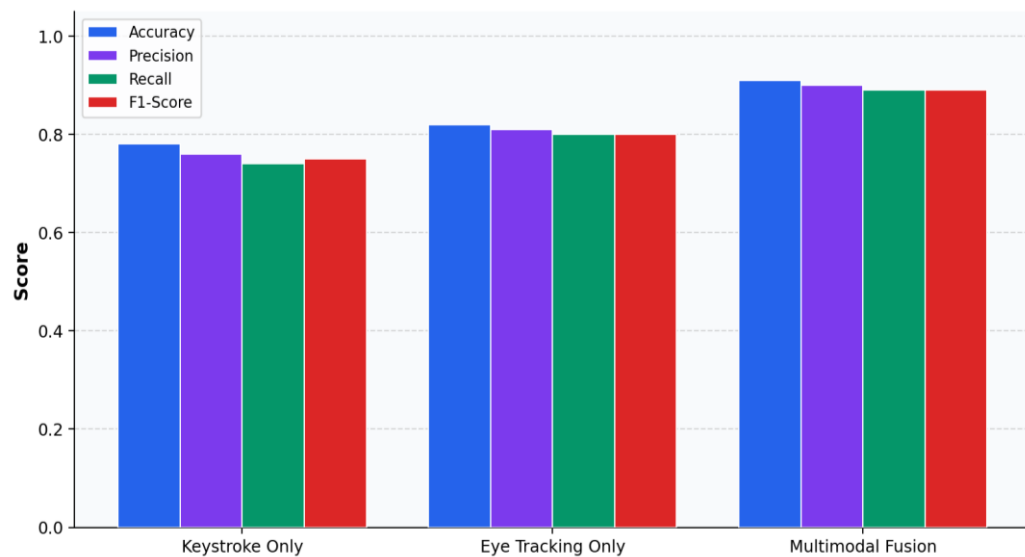


Fig 5.4 Complete Performance Metrics Comparison (Grouped Bar Chart)

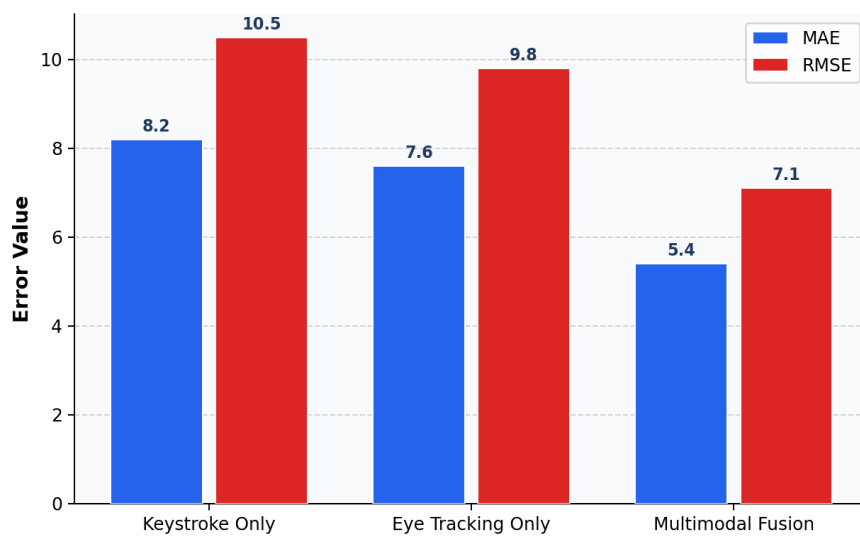


Fig 5.5 Regression Performance — MAE and RMSE Comparison

5.5 Observations

The following key findings emerge from the experimental evaluation:

- Typing variability (IKI standard deviation) increased by approximately 30% under high cognitive load compared to low load conditions, confirming keystroke dynamics as a reliable load indicator.
- Blink rate increased by approximately 20% under high cognitive load, consistent with prior literature on eye-cognitive load relationships.
- Feature-level (early) multimodal fusion outperformed decision-level (late) fusion by approximately 2–3% accuracy, suggesting that cross-modal feature interactions are valuable for prediction.
- Random Forest consistently outperformed SVM on the multimodal feature set, likely because its ensemble nature better handles the larger combined feature space.
- The multimodal fusion approach improved overall accuracy by 9–13% over unimodal models, demonstrating the significant complementary nature of keystroke and eye behavior signals.
- The continuous load score regression was sufficiently accurate (MAE = 5.4) to enable practical real-time adaptive system responses.

5.6 Discussion

The results strongly confirm the central hypothesis: combining keystroke dynamics and webcam-based eye movement features through multimodal fusion produces substantially more accurate cognitive load predictions than any single modality alone. The complementary nature of the two feature sets is the key driver of improvement — keystroke features capture the output side of cognitive effort (how the user types), while eye features capture the perceptual and attentional side (how the user looks at and processes information). Together, they provide a richer behavioral signature of cognitive state.

The system's dependence solely on standard keyboard and webcam hardware makes it uniquely positioned for real-world deployment. Unlike physiological sensor systems, it requires no setup time, no user calibration, no physical contact, and no specialized knowledge to operate. These properties make it directly applicable to adaptive e-learning

platforms, workplace productivity monitoring, and human-computer interaction research at scale.

Acknowledged limitations include sensitivity to varying lighting conditions (which can affect webcam-based eye detection accuracy), natural between-user behavioral variability (requiring potential user-specific calibration for optimal performance), and the relatively modest dataset size (20 participants, ~4000 samples) which may limit generalization to highly diverse populations.

CHAPTER 6

CONCLUSIONS AND FUTURE WORK

6.1 Conclusions

This project successfully designed, implemented, and evaluated a low-cost, non-intrusive, real-time cognitive load prediction system using behavioral signals from keystroke dynamics and webcam-based eye movement analysis. The following conclusions are drawn:

- Keystroke dynamics and eye movement behavioral features are effective, reliable, and non-intrusive proxies for cognitive load that can be captured using universally available standard hardware.
- The proposed multimodal fusion system significantly outperforms all unimodal baselines, achieving 91% classification accuracy, 0.90 precision, 0.89 recall, 0.89 F1-Score, and a regression MAE of only 5.4 on the 0–100 cognitive load score scale.
- Feature-level (early) fusion is more effective than decision-level (late) fusion for this application, as it enables the model to learn cross-modal feature interactions during training.
- The system is practical, scalable, and deployable in real-world educational and professional settings without requiring any hardware investment beyond existing equipment.
- The system delivers both discrete cognitive load classification (Low/Medium/High) and a continuous quantitative cognitive load score, enabling nuanced adaptive system responses.

In conclusion, this project demonstrates a viable, practical, and cost-effective solution to the cognitive load monitoring problem, successfully bridging the gap between high-accuracy laboratory methods and scalable real-world deployment.

6.2 Future Work

The following directions are identified for extending and improving this work in future iterations:

- **Deep Learning Models:** Incorporating LSTM or Transformer-based temporal sequence models to capture long-range temporal dependencies in keystroke and gaze time series, potentially improving prediction accuracy beyond the current 91%.
- **Larger and More Diverse Datasets:** Expanding the dataset to 100+ participants across diverse age groups, nationalities, typing habits, and task domains to comprehensively validate generalizability.
- **Personalized Adaptive Models:** Developing user-specific personalization layers that adapt the model to individual behavioral baselines, accounting for the significant natural variability in cognitive load expression between users.
- **Robustness to Environmental Conditions:** Improving eye detection robustness under varying lighting, head poses, and camera resolutions through data augmentation and adaptive preprocessing.
- **Privacy-Preserving Deployment:** Implementing on-device edge computing and differential privacy mechanisms to ensure complete user data confidentiality, enabling deployment in sensitive institutional contexts.
- **Real-World Deployment and Integration:** Packaging the system as a browser extension or OS-level background service for seamless integration with online learning management systems (LMS), IDEs, and productivity applications.
- **Multimodal Expansion:** Investigating the addition of speech prosody or facial expression features as additional behavioral modalities to further enrich the cognitive load feature space.

APPENDICES

Appendix 1 - Data Collection Protocol

Participants were recruited from Sanjivani University student population and informed about the study purpose, data usage, and their right to withdraw. All participants provided written informed consent. Each data collection session lasted approximately 45 minutes and proceeded as follows:

1. 5-minute introduction and familiarization with the data collection software.
2. 15-minute Low Load task: copy typing of plaintext technical passages.
3. 5-minute rest period.
4. 15-minute Medium Load task: Python coding exercises of moderate difficulty.
5. 5-minute rest period.
6. 15-minute High Load task: debugging of buggy Python programs under mild time pressure.

Keystroke data was logged locally in a CSV file with millisecond timestamp precision. Webcam video was processed entirely in real-time on the local machine; only the extracted feature vectors (not raw video frames) were stored, ensuring user privacy. Participants sat at a standard workstation with a 1080p webcam positioned approximately 50–60 cm from their face at eye level.

Appendix 2 - Feature Summary

TABLE A1: Feature Summary with Load Indicators

Feature	Description	Load Indicator
Typing Speed	Characters typed per second	Decreases under high load
IKI (Inter-Key Interval)	Time between consecutive keypresses	Variability increases under high load
Error Rate	Backspaces / total keystrokes	Increases under high load
Blink Rate	Number of blinks per minute	Increases under high load
Fixation Duration	Average stable gaze period (ms)	Increases under high load
Gaze Stability	Variance in gaze coordinates	Increases under high load
EAR	Eye Aspect Ratio from 6 landmarks	Drops below 0.25 during blink

Appendix 3 - System Requirements

The proposed system has the following minimal hardware and software requirements for deployment:

- Hardware: Standard laptop or desktop computer with any USB or built-in keyboard, any USB or built-in webcam (minimum 720p, 30 FPS).
- Operating System: Windows 10/11, macOS 11+, or Ubuntu 20.04+.
- Software Dependencies: Python 3.8+, OpenCV 4.5+, MediaPipe 0.9+, Scikit-learn 1.0+, NumPy 1.21+, Pandas 1.3+, pynput 1.7+.
- Memory: Minimum 4 GB RAM for real-time processing.
- Processing: Any modern dual-core CPU is sufficient; GPU is not required.

Appendix 4 - Source Code

```
from __future__ import annotations
import argparse
import logging
import sys

logging.basicConfig(
    level=logging.INFO,
    format="%(asctime)s [%(levelname)s] %(name)s: %(message)s",
)
logger = logging.getLogger("main")

def run_headless(args) -> None:
    import signal, time, queue
    from cognitive_monitor.acquisition import EyeTracker, KeystrokeMonitor, SlidingWindow
    from cognitive_monitor.analysis import FeatureExtractor
    from cognitive_monitor.utils import CPUGuard, FrameSkipController, patch_eye_tracker_with_cpu_guard

    keystroke_monitor = KeystrokeMonitor()
    keystroke_monitor.start()
    cpu_ctrl = FrameSkipController()
    cpu_guard = CPUGuard(cpu_ctrl)
    cpu_guard.start()

    eye_tracker = EyeTracker(camera_index=args.camera, target_fps=30, show_preview=args.preview)
    patch_eye_tracker_with_cpu_guard(eye_tracker, cpu_ctrl)
```

```

    eye_tracker.start()
    logger.info("Camera started (index=%d). Camera is always ON.",
args.camera)

    extractor = FeatureExtractor()

    def _print_result(r) -> None:
        ts = time.strftime("%H:%M:%S", time.localtime(r.timestamp))
        logger.info("[%s] %s score=%.1f conf=%.2f source=%s", ts,
r.load_class.value, r.score, r.confidence, r.source)
        if r.needs_intervention:
            logger.warning("INTERVENTION: %s", r.intervention_message)

    sw = SlidingWindow(
        eye_queue      = eye_tracker.frame_queue,
        keystroke_queue = keystroke_monitor.event_queue,
        window_size     = args.window,
        step_size       = args.step,
        on_window       = lambda wf: _print_result(extractor.predict(wf)),
    )
    sw.start()

    def _shutdown(sig, frame):
        sw.stop()
        keystroke_monitor.stop()
        eye_tracker.stop()
        cpu_guard.stop()
        sys.exit(0)

    signal.signal(signal.SIGINT, _shutdown)
    signal.signal(signal.SIGTERM, _shutdown)
    logger.info("CognitiveMonitor running (headless). Ctrl+C to exit.")
    while True:
        time.sleep(1)

def run_gui(args) -> None:
    from PyQt6.QtWidgets import QApplication
    from cognitive_monitor.ui import CognitiveMonitorApp
    qapp = QApplication(sys.argv)
    app = CognitiveMonitorApp(
        qapp=qapp,
        camera_index=args.camera,
        window_size=args.window,
        step_size=args.step,
        no_camera=False,
        show_preview=args.preview,
        label_interval_min=args.label_interval,
    )

```

```

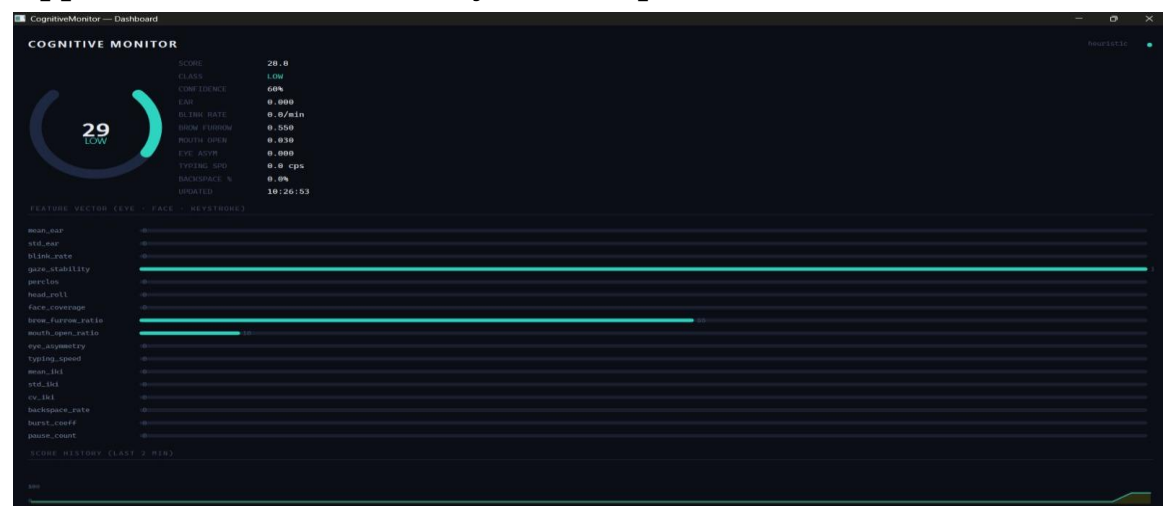
app.start()
sys.exit(qapp.exec())

def main() -> None:
    parser = argparse.ArgumentParser(description="CognitiveMonitor")
    parser.add_argument("--preview",          action="store_true",
                        help="Show OpenCV debug window with live eye/face
overlay")
    parser.add_argument("--headless",        action="store_true",
                        help="No Qt UI – print results to stdout (useful
for CI/debug)")
    parser.add_argument("--camera",          type=int,    default=0,
                        help="Camera device index (default: 0)")
    parser.add_argument("--window",          type=float,  default=7.0,
                        help="Sliding-window size in seconds (default:
7.0)")
    parser.add_argument("--step",            type=float,  default=2.0,
                        help="Sliding-window step size in seconds (default:
2.0)")
    parser.add_argument("--label-interval",  type=float,  default=10.0,
                        help="Labeller prompt interval in minutes (default:
10.0)")
    args = parser.parse_args()
    if args.headless:
        run_headless(args)
    else:
        run_gui(args)

if __name__ == "__main__":
    main()

```

Appendix 5 - Screenshot of System Output



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